

Alfredo Bautista Ríos

Computer Systems Engineer (Recent Graduate)

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Professional Profile

Computer Systems Engineer motivated by building technological solutions that integrate data, software, and artificial intelligence to solve real-world problems. I am passionate about understanding systems in depth — from architecture to implementation — and transforming ideas into functional and maintainable tools. I aim to grow in environments where I can combine data, software development, and open-source principles to deliver scalable and meaningful impact.

Education

ESCOM – IPN — B.Sc. in Computer Systems Engineering

Aug 2021 – Jan 2026

Strong academic background in data structures, algorithms, databases, software engineering, and artificial intelligence. Relevant coursework in natural language processing, computer vision, image analysis, and native mobile applications. Active member of the competitive programming club focused on algorithmic problem solving. Academic experience in implementing and comparing classical machine learning models, as well as developing image analysis applications with graphical user interfaces.

Thesis Project — Quiroscopio.

Wearable gesture recognition system based on a microcontroller and inertial sensors. Developed embedded firmware, implemented Bluetooth Low Energy communication, designed software for data acquisition and processing, and trained a machine learning model for gesture classification. Integrated output with a Linux-based system to trigger automated actions.

CECyT 3 IPN — Technical Degree in Computer-Aided Manufacturing

Aug 2017 – Jun 2021

Technical education with emphasis on automation, industrial processes, and applied engineering fundamentals, with strong academic performance.

Professional Experience

BI Trainee — Profuturo S.A. de C.V.

Feb 2025 – Present

Contributed to the design and redesign of data models and mappings for multiple business units, using SAS as the primary platform. Developed ETL processes, automation workflows, dashboards, and SQL queries across different data volume scenarios. Integrated and consumed APIs within analytical and operational workflows. Maintained direct communication with technical and non-technical stakeholders, handling requirement analysis, documentation, and process reengineering. Hands-on experience with Databricks, Python, SQL, and Power BI.

Relevant Coursework

Samsung Innovation Campus (UDEM) — Artificial Intelligence and Leadership

Foundations of artificial intelligence, mathematics for data science, exploratory data analysis, and machine learning. Complementary training in technical leadership, teamwork, and effective communication in collaborative environments.

Additional certifications in cloud computing fundamentals (Google Cloud Computing Foundations), Linux system administration (LPI Linux Essentials), and English proficiency at B2 level.

Technical Skills

Programming Languages: C/C++, Python, SQL, PHP, Bash

Development: Linux, GitHub, HTML, graphical interfaces, background services (daemons)

Methodologies: Scrum, Kanban, Waterfall

Tools: SAS, Databricks, Power BI

Systems: Raspberry Pi, Arduino

Data & BI: ETL pipelines, data modeling, exploratory data analysis, dashboards

Machine Learning: supervised and unsupervised models, feature engineering, model training and evaluation using formal metrics

Core Competencies: technical communication, requirements analysis, stakeholder negotiation, collaborative leadership, and continuous self-learning.